

CompSci 234/NetSys 210
Advanced Topics in Networking
Winter 2019

Unit 7: DASH and RTP Streaming

1

Cheng-Hsin Hsu (chsu@cs.nthu.edu.tw)

Slide adopted from Profs. Ooi and Zimmerman, and Timmerers' materials

Agenda

- DASH Overview
- ABR: Adaptive Bitrate Allocation
- WebRTC and Underlying Protocols (RTP, RTCP, RTSP)
- DASH versus RTP

Where Did DASH Come From?

■ Video not accessible

- Behind a firewall
- Plugin not available
- Bandwidth not sufficient
- Wrong/non-trusted device
- Wrong format

■ Fragmentation

- Devices
- Content Formats
- DRMs

■ Low Quality of Experience

- Long start-up delay
- Frequent Re-buffering
- Low playback quality
- No lip-sync
- No DVD quality (language, subtitle)

■ Expensive

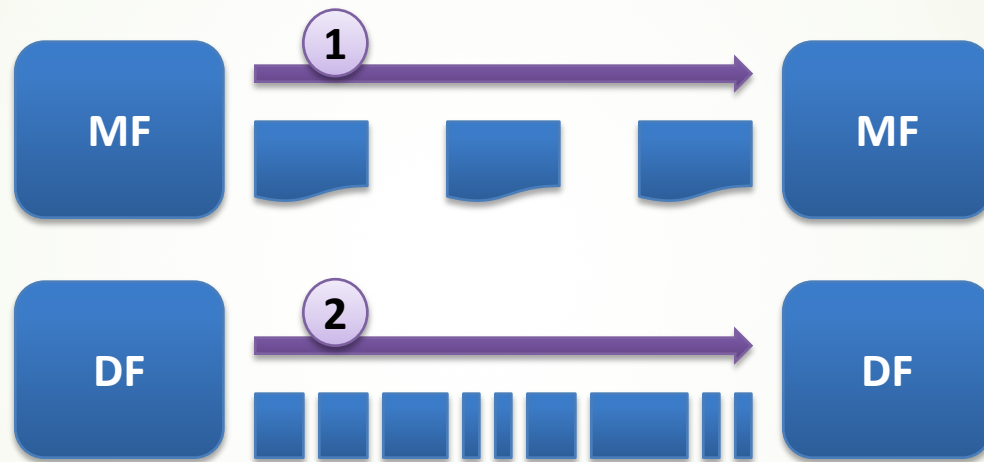
- Sucks my bandwidth
- Need a dedicated devices
- Other costs...

To resolve the user frustration ⇨ Open Standards

HTTP Streaming of Media

Server

Client



ISOBM
FF

easy
conversion

M2TS

ISOBM
FF

easy
conversion

M2TS

ISOBMFF ... ISO Base Media File Format (e.g., mp4 – others: avi)

M2TS ... MPEG-2 Transport Stream (e.g., DVB, DMB)

MF ... Manifest Format (e.g., MPD, FMF)

DF ... Delivery Format (e.g., F4F, 3gs)

(Earlier) Proprietary Solutions



HTTP Live Streaming



Microsoft
Silverlight

Smooth Streaming



Adobe HTTP Dynamic
Streaming

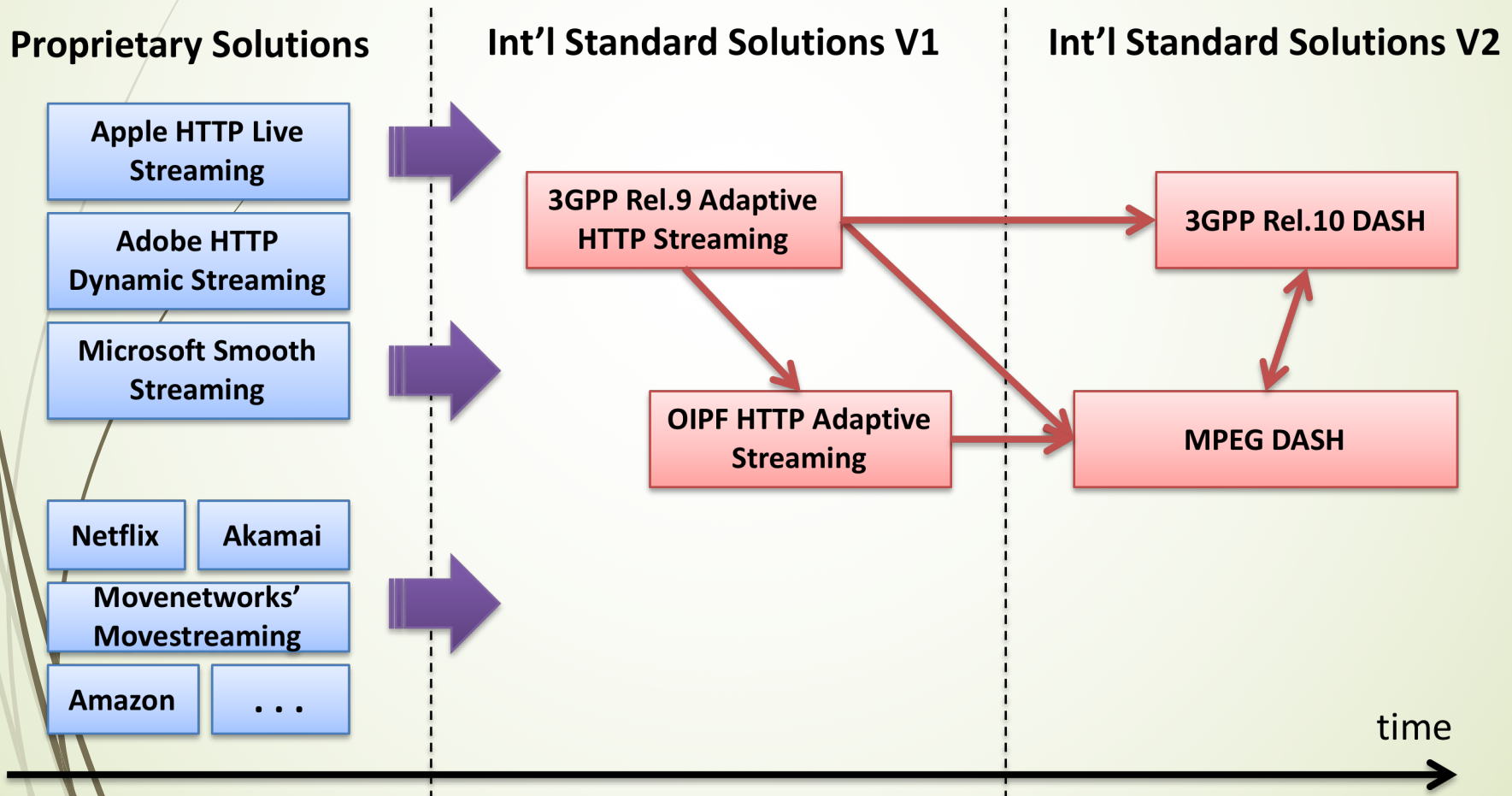
Combined A/V
streams only



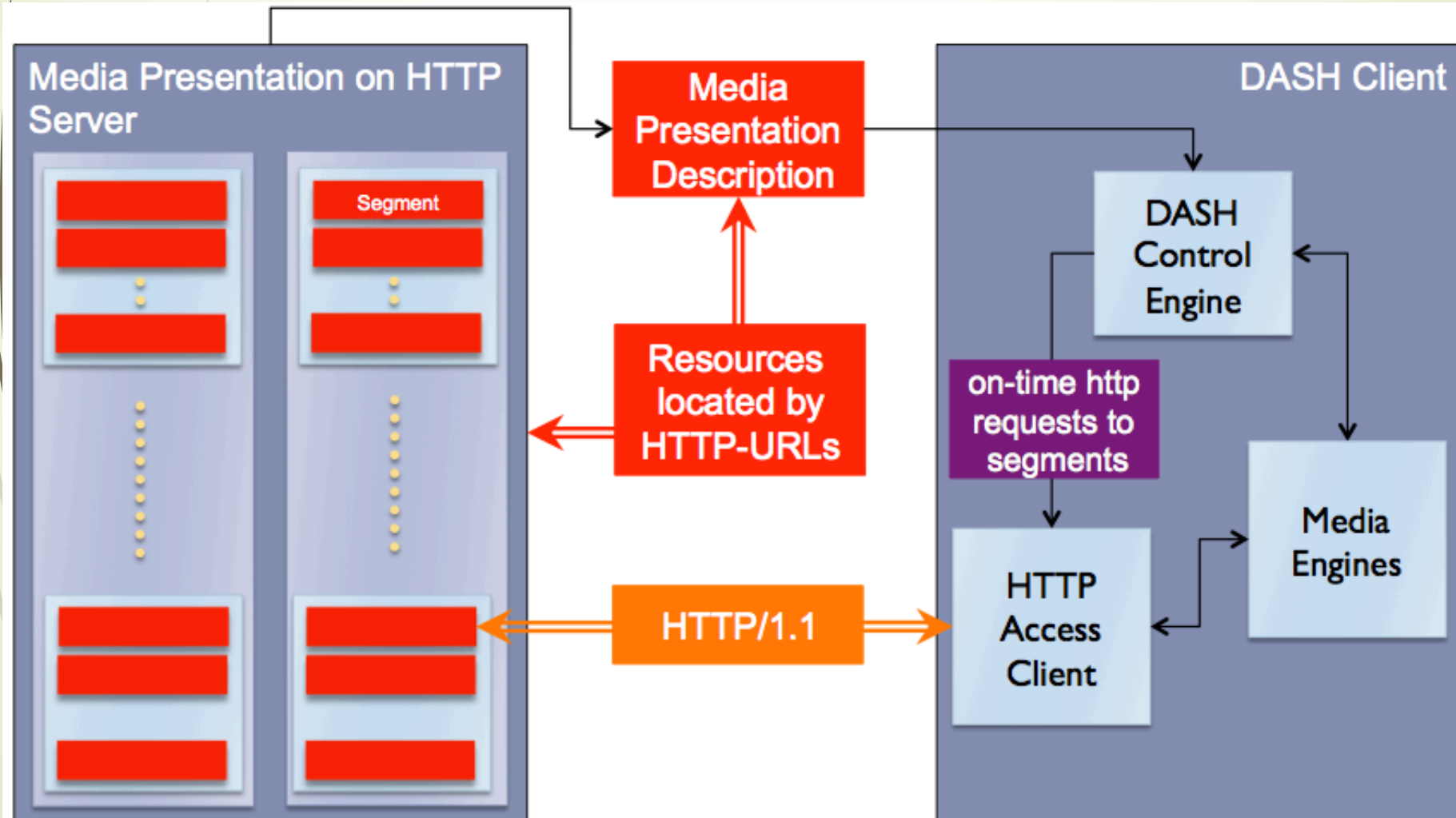
MPEG DASH

Separate Audio/Video

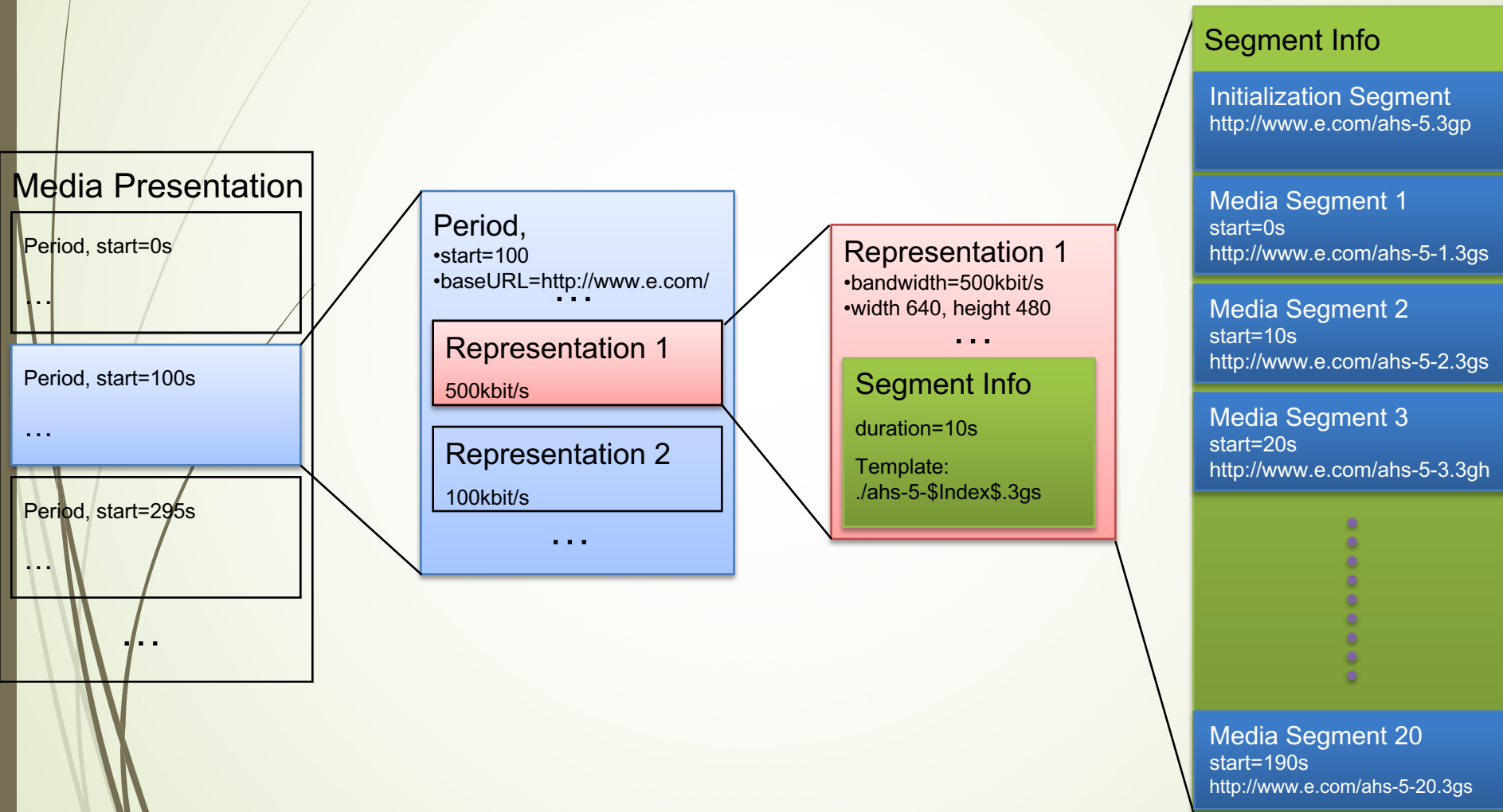
Dynamic Adaptive Streaming over HTTP (DASH)



What Are (Aren't) Specified in the Standard



DASH Data Model



Media Presentation Description (MPD)

- Info for choosing groups or representations
 - codec, DRM, language, resolution, bandwidth
- **Access and Timing Information**
 - HTTP-URL(s) and byte range for each accessible Segment
 - Earliest next update of the MPD on the server
 - Segment availability start and end time in wall-clock time
 - Approximated media start time and duration of a Media Segment in the media presentation timeline
 - For live service, instructions on starting playout such that media segments will be available in time for smooth playout in the future
- Info for switching across representations

DASH Groups and Subsets

Group id="grp-1"

Representation id="rep-1"

Representation id="rep-2"

...

Representation id="rep-n"

Group id="grp-2"

Representation id="rep-1"

Representation id="rep-2"

...

Representation id="rep-n"

...

Group by codec, language, resolution, bandwidth, views, etc. – very flexible (in combination with xlink)!

- Ranges for the @bandwidth, @width, @height and @frameRate

Subset id="ss-1"

Contains group="grp-1"

Contains group="grp-4"

Contains group="grp-7"

Subsets

- Mechanism to restrict the combination of *active* Groups
- Expresses the intention of the creator of the Media Presentation

Segment Index

11

- Hierarchical binary info on
 - Accessible units of data in a media segment
 - Each unit is described by
 - Byte range in the segments (easy access through HTTP partial GET)
 - Accurate presentation duration (seamless switching)
 - Presence of representation access positions, e.g. IDR frames
- Provides a compact bitrate-over-time profile to client
- Generic Data Structure usable for any media segment format, e.g. ISO BMFF, MPEG-2 TS, MP4, and etc.

Different Segment Index Approaches

Segment Index in MPD only

```
<MPD>
```

```
...
```

```
<URL sourceURL="seg1.mp4"/>
```

```
<URL sourceURL="seg2.mp4"/>
```

```
</MPD>
```

seg1.mp4

seg2.mp4

...

```
<MPD>
```

```
...
```

```
<URL sourceURL="seg.mp4" range="0-499"/>
```

```
<URL sourceURL="seg.mp4" range="500-999"/>
```

```
</MPD>
```

seg.mp4

Segment Index in MPD + Segment

```
<MPD>
```

```
...
```

```
<Index sourceURL="idx.mp4"/>
```

```
<URL sourceURL="seg.mp4"/>
```

```
</MPD>
```

idx.
mp4

seg.mp4

Segment Index in Segment only

```
<MPD>
```

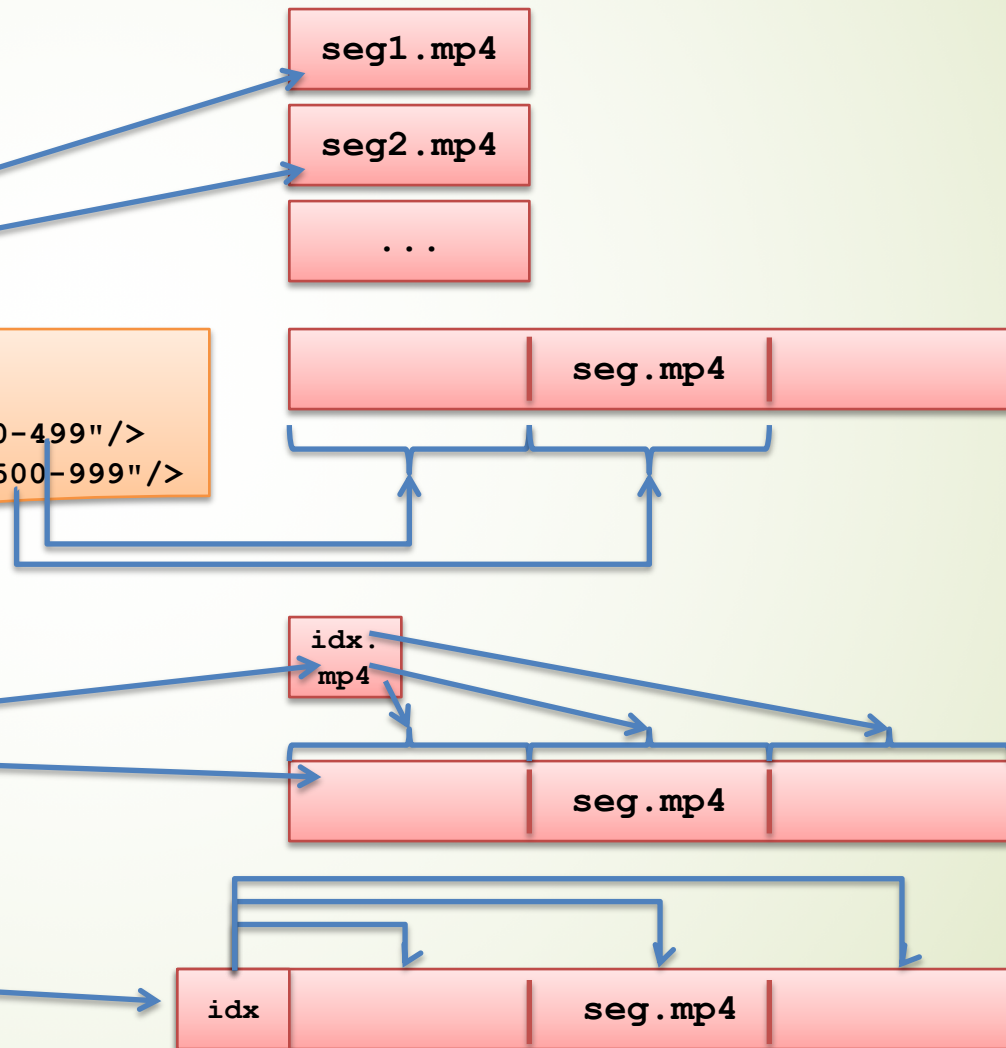
```
...
```

```
<BaseURL>seg.mp4</BaseURL>
```

```
</MPD>
```

idx

seg.mp4



Summary: DASH Streaming

- Asynchronous delivery of the same content to many users is a first-class network service
 - HTTP CDNs may not be the “perfect” architecture, but it’s working pretty well at scale
 - Naturally traverse through NAT, Firewall, and Gateway boxes
- DASH is simple: **Fits market needs!**
- **Some parameters/designs need to be carefully chosen/done**

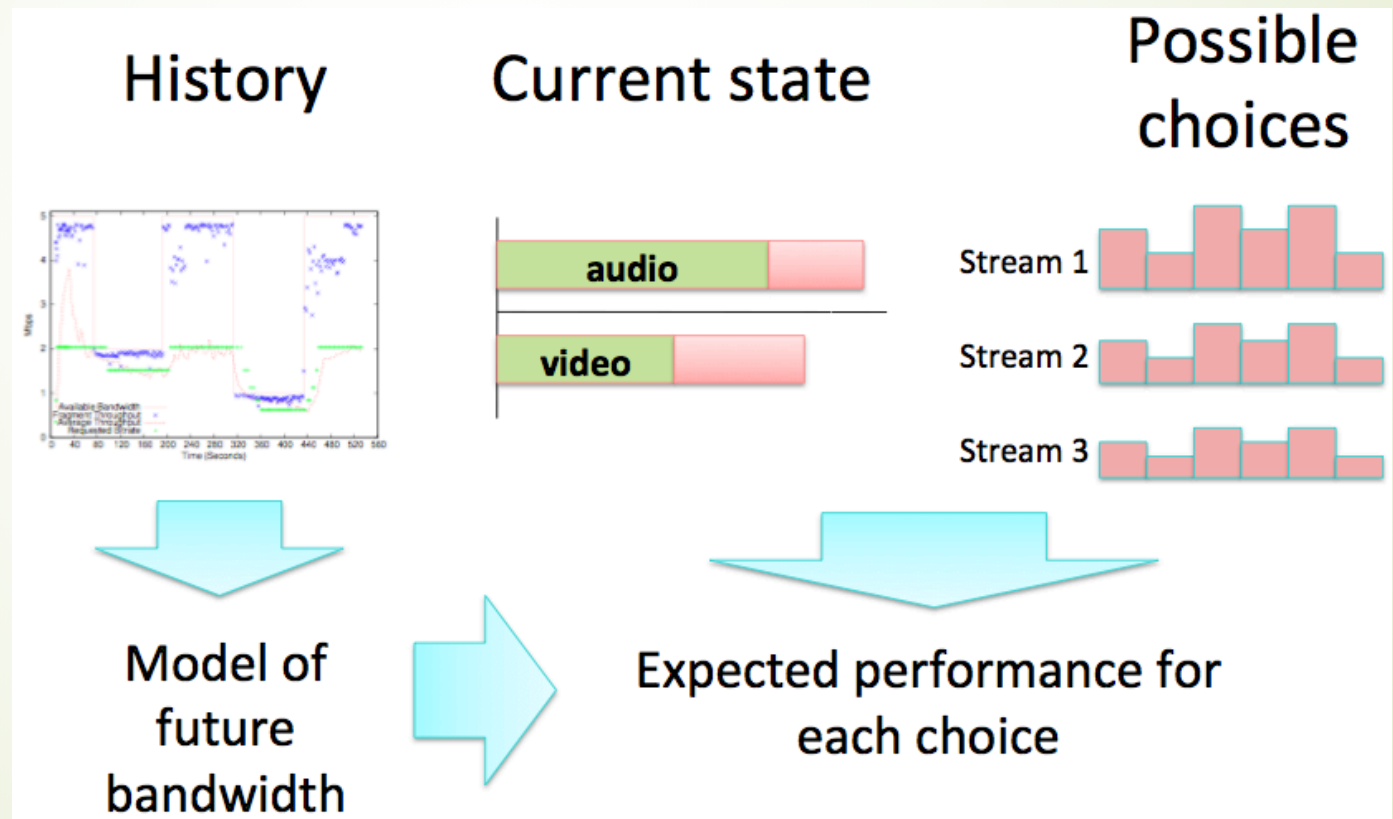
Segment duration	Advantages	Disadvantages
Short	• Commonality with Live • High switching granularity on segment level	• Large number of files • Large number of URLs • Fixed request size • switching granularity on segment level
Long	• Small number of files • Small number of URLs • High switching granularity • Flexible request sizes • Improved cache performance	• Need for Segment Index • Difference from Live

Agenda

- DASH Overview
- ABR: Adaptive Bitrate Allocation
- WebRTC and Underlying Protocols (RTP, RTCP, RTSP)
- DASH versus RTP

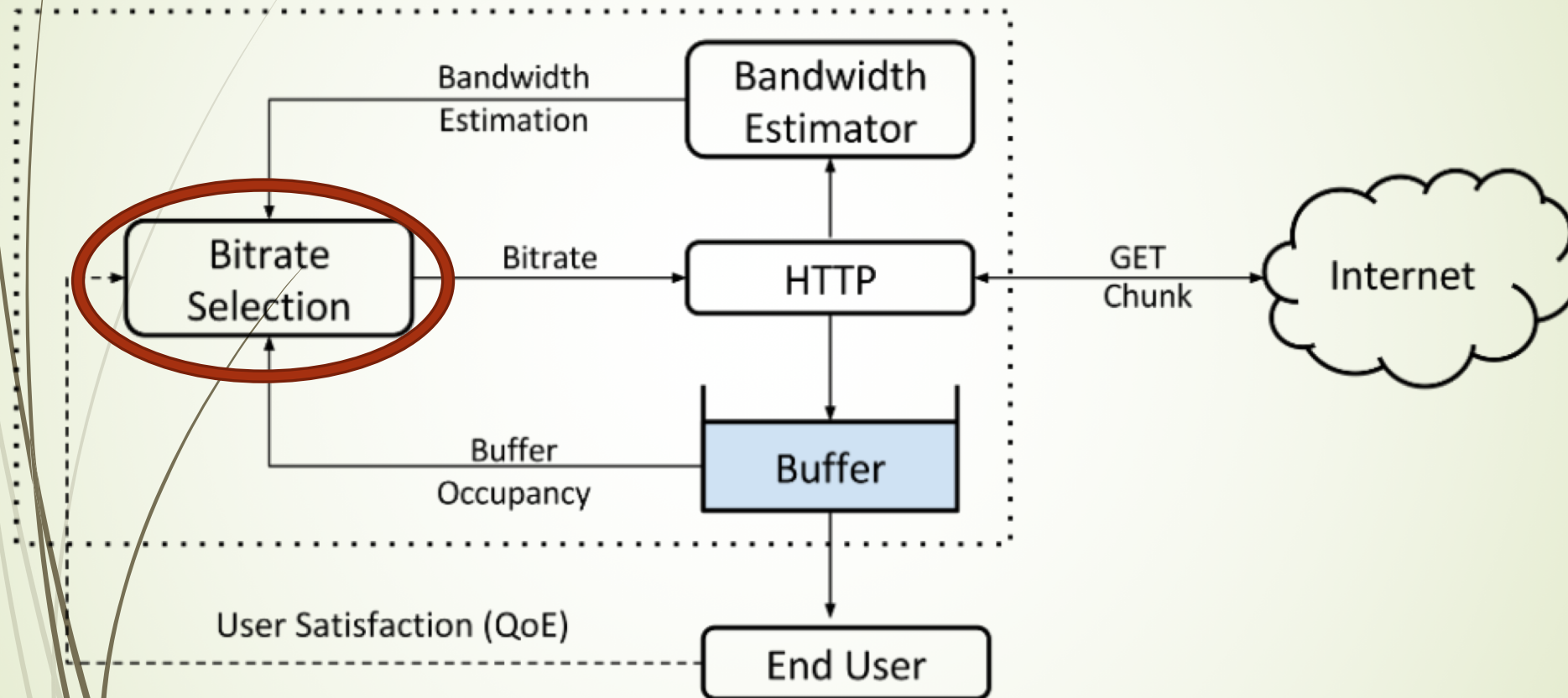
Rate Adaptation Problem

Pick segments and request timing to minimize re-buffering events and maximize quality



Q: Who (server or client) is making this decision?

Client Side Bit Rate Adaptation



Q: What are the KEY inputs to the bitrate adaptation algorithms?

Two Main Approaches and Potential Issues

- Bandwidth based algorithms
 - Bandwidth estimation is **inherently hard**
 - Each client works alone => unfairness, congestion
 - Heterogeneous devices with various capabilities?
 - May suffer from buffer starvation and low QoE
- Buffer-level based algorithms
 - Frequent bitrate switch → low QoE
 - Slower reactions to rapid/sudden bandwidth changes
- **Other approaches?**

Other Adaptive Bitrate Allocation Approaches

- Server side approaches
 - Centralized decision at the streaming servers
- In-network adaptations (with MANE)
 - Some intermediate network equipment participate in the adaptations
 - E.g., if the downstream network is congested why bother to send the segment which will be late anyway?
- Other tools that have been thrown in
 - Scalable video coding (SVC): Layered Coding or Multiple Description Coding
 - Software-Defined Networking

Agenda

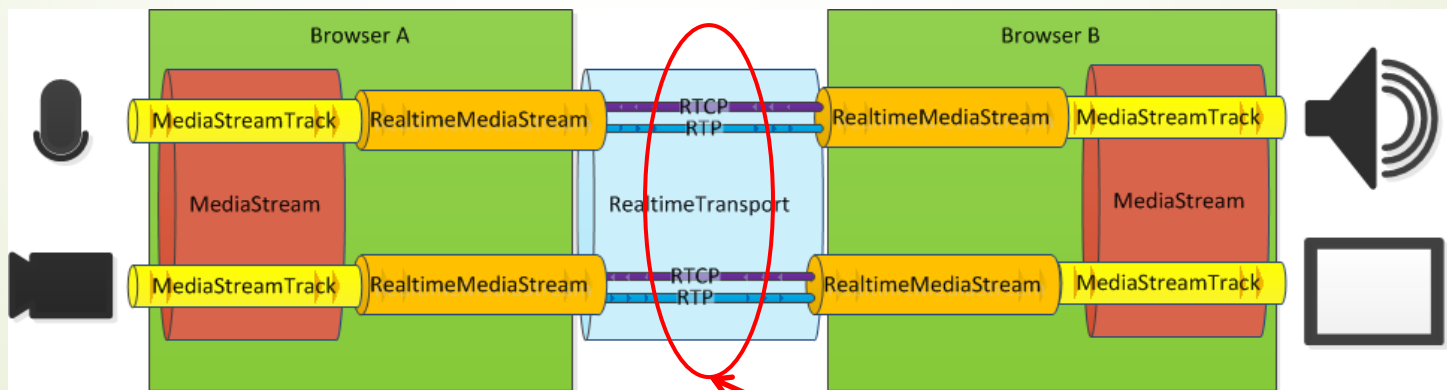
- DASH Overview
- ABR: Adaptive Bitrate Allocation
- WebRTC and Underlying Protocols (RTP, RTCP, RTSP)
- DASH versus RTP

Limitations of DASH Streaming

- DASH has a number of advantages
 - Server is simple, i.e., regular web server
 - No firewall problems (use port 80 for HTTP)
 - Web caching works (this is very tricky for UDP, **Why?**)
- However, DASH sends **segments**, typically 2-10 sec long
- DASH client has to buffer a few segments, and does **not**:
 - Provide low latency for interactive, two-way applications (e.g., video conferencing)
- **Presentational versus conversational video services**

Alternate Solution: WebRTC

- Web browsers with Real-Time Comm. (RTC) capabilities via simple JavaScript APIs
- Pipeline for video conferencing in WebRTC (only one-way shown):



RTP/RTCP